## Formula Sheet

z-score for probability distributions

$z = \frac{x-\mu}{\sigma}$

test-statistic for sampling distributions

$z = \frac{\bar{x}-\mu_o}{\sigma/\sqrt{n}}$ (rare: if we know $\sigma$, but not $\mu$)

$z = \frac{\hat{p}-\pi_o}{\sqrt{\frac{\pi_o(1-\pi_o)}{n}}}$

$t = \frac{\bar{x}-\mu_o}{s/\sqrt{n}}$

Confidence intervals

$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$ (rare: if we know $\sigma$, but not $\mu$)

$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$

$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$